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$$f(u, v) = (2u^2 + 3v^2) \cdot e^{-u^2-v^2}$$

$$\begin{aligned} Df(u, v) &= \left((4u + 2u^2 \cdot (-2u) + 3v^2 \cdot (-2u)) \cdot e^{-u^2-v^2} \quad (2u^2 \cdot (-2v) + 6v + 3v^2 \cdot (-2v)) \cdot e^{-u^2-v^2} \right) \\ &= \left((-4u^3 + 4u - 6uv^2) \cdot e^{-u^2-v^2} \quad (-6v^3 + 6v - 4u^2v) \cdot e^{-u^2-v^2} \right) \end{aligned}$$

References